Spatial Pressure Theory: A Gravitational Origin Mechanism Based on Uniform Background Field Perturbation

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Abstract

Newton's law of universal gravitation and Einstein's general relativity have achieved high-precision mathematical descriptions of gravitational phenomena, yet neither has directly addressed the question of why gravity exists—that is, the physical origin of gravity. This paper proposes the **Spatial Pressure Theory (SPT)**. Its core proposition is as follows: cosmic space is not an absolute void. The vacuum is a background with stable spatial extension, within which a continuous, invisible substrate medium, termed the **Spatial Pressure Medium**, is uniformly distributed. The equivalent mechanical property of this medium is spressure—uniformly distributed and imperceptible in the absence of matter. When matter occupies a spatial region, the Spatial Pressure Medium in that region is displaced, forming a localized density defect that creates a non-uniform spressure field. **Gravity is simply spressure with a gradient. Uniform spressure corresponds to the absence of gravity; spressure with a gradient is gravity itself.** As a static continuous medium, the Spatial Pressure Medium ensures through its elastic response capability that pressure disturbances propagate dynamically outward as **gravitational waves**; its continuity ensures that the pressure gradient extends smoothly into the distance, accounting for the **long-range nature of gravity**. Without altering the observational predictions of existing gravitational theories, SPT provides a unified medium-dynamical explanation for gravitational phenomena and offers falsifiable observational predictions through pressure saturation effects in extreme strong-field environments.

Key words: Spatial Pressure Theory, SPT, Spatial Pressure Medium, spressure, gravitational origin, density defect, pressure gradient, gravitational waves, long-range nature

1 Introduction

Gravity is one of the four fundamental interactions in nature. Newton's law of universal gravitation defines gravity as a long-range interaction between masses, achieving quantitative calculations at macroscopic scales through the inverse-square law [1]. Einstein's general relativity reinterprets gravity as a manifestation of spacetime curvature, establishing a deep connection between gravity and the structure of spacetime [2]. Both theories have passed rigorous experimental verification within their respective domains of applicability—from terrestrial free-fall to planetary orbits, from gravitational redshift to light deflection, from the precession of Mercury's perihelion to gravitational waves—each high-precision observation has reinforced this magnificent mathematical edifice.

Yet beneath this edifice, the most fundamental question remains unanswered: **Why does gravity exist?**

Newtonian mechanics does not explain how mass generates gravity. General relativity does not clarify what the physical carrier of spacetime curvature is. Existing theories are essentially mathematical modeling and equivalent descriptions of gravitational phenomena, rather than a trace back to their physical origin. A theory can precisely tell you how to calculate gravity, yet cannot tell you where gravity comes from. This is perhaps the deepest silence in modern physics.

This paper proposes the **Spatial Pressure Theory (SPT)**, seeking to provide a unified medium-dynamical explanation for gravity without altering the observational predictions of existing theories, thereby bridging the gap between phenomenological fitting and origin tracing. It should be noted that this paper serves as the mechanistic exposition and conceptual foundation for SPT. Its core task is to establish a clear, complete physical picture and a medium-dynamical logical framework. This version deliberately sets aside complex tensor and partial differential field equation systems, prioritizing the qualitative mechanical language of classical continuum mechanics and topological defect theory for logical construction. In the history of science, a clear physical picture often precedes mathematical formalization. The multi-scale equivalent compatibility demonstrated by this model—the mechanical isomorphism with Newtonian mechanics, general relativity, and the Casimir effect—has passed the test of qualitative self-consistency. More specific quantitative field equations and numerical simulations will be undertaken as independent subsequent projects.

2 The Spatial Pressure Medium: A Continuous Medium Distributed in the Vacuum Background

2.1 Existence Form and Core Characteristics

This paper proposes a core hypothesis: cosmic space is not an absolute void. The vacuum is a background with stable spatial extension—it does not fluctuate, does not warp, and is not compressed, providing merely the spatial extension for all physical existence. Within this vacuum background, a continuous, invisible substrate medium is uniformly distributed, defined as the **Spatial Pressure Medium**.

The key feature distinguishing this medium from the historical concept of aether is that it is not a material medium composed of known elementary particles, but a yet-unidentified, pre-physical continuous medium filling the vacuum background. It is not any known field in the Standard Model, nor is it directly equivalent to any entity within the Standard Model of particle physics. The Spatial Pressure Medium differs from the vacuum energy in quantum field theory—vacuum energy is the ground state fluctuation of quantum fields, existing within spacetime; the Spatial Pressure Medium is the substrate medium pervading the vacuum, within which quantum fields themselves propagate.

In this model, the equivalent mechanical property exhibited by the Spatial Pressure Medium is spressure. The former emphasizes its setting as the vacuum-filling medium; the latter emphasizes its pressure characteristic that can participate in mechanical descriptions. The two are treated as two equivalent descriptions of the same entity in the mathematical treatment of this model.

The Spatial Pressure Medium possesses the following basic physical properties:

1. Universal Carrier: It is the propagation medium for light, particles, electromagnetic waves, and quantum fields. The argument for this property runs as follows: light, electromagnetic waves, gravitational waves, and particles all propagate through the vacuum—a rigorously verified observational fact. This model is grounded in the premise that absolute nothingness cannot serve as a carrier of wave propagation. It follows that the so-called vacuum cannot physically be absolute nothingness; it must be filled by some material medium capable of carrying waves. This paper defines this medium as the Spatial Pressure Medium. Its existence is not an ad hoc hypothesis introduced to explain gravity, but a logical necessity arising from the physical inquiry into why the vacuum possesses the capacity to carry waves. A closed loop is thus formed: the vacuum carries waves (observational fact) → this model holds that absolute nothingness cannot serve as a carrier of wave propagation (modeling premise) → the vacuum must be filled by some physical medium (logical necessity) → this medium is defined as the Spatial Pressure Medium (theoretical naming) → the uniform state of this medium manifests as the gravity-free vacuum (physical picture closure).

2. Not Directly Detectable: The Spatial Pressure Medium is too uniform and continuous, and does not scatter or reflect off known particles, making it impossible to directly observe its existence through conventional experimental means. It is not nothing, but an invisible medium.

3. Indirectly Verifiable: Although the Spatial Pressure Medium cannot be directly detected, its existence is indirectly manifested through observable mechanical effects. The universal existence of gravity itself is the most direct evidence: if the vacuum were absolute nothingness, it would be impossible to explain how gravity is transmitted between objects. If gravity is the resultant force effect of the spressure gradient, then every gravitational observation—from terrestrial free-fall to planetary orbits, from gravitational waves to galactic rotation curves—is an indirect verification of the medium's existence.

4. Uniformity: In the absence of matter perturbations, the Spatial Pressure Medium is uniformly distributed throughout space in density, representing the lowest energy state of the universe.

5. Isotropy: The physical properties at any point in space do not vary with direction, satisfying spherical symmetry conditions.

6. Equilibrium Stability: At macroscopic scales, it is always in a state of thermodynamic equilibrium, with no spontaneous perturbations.

7. No Net Force Effect: In the equilibrium state, spressure is uniformly distributed; the pressure exerted on any object in space from all directions cancels out, exhibiting no observable mechanical effect.

8. Static and Non-flowing: The Spatial Pressure Medium is a static, non-flowing continuous medium. The motion of matter within it is not a displacement that creates circulation or drag, but rather the spatial displacement of a defect state within a stationary medium. This static medium serves only as a reference configuration for theoretical modeling. Because it does not participate in the dynamical drag of matter and its coupling mode satisfies the principle of locality-preserving response, this model does not introduce an observable preferred reference frame at the observable scale. Its low-energy effective theory remains consistent with the Lorentz covariance of special relativity.

9. Finite Compressibility and Local Displacement: Although the Spatial Pressure Medium can transmit waves at microscopic scales, it behaves as an incompressible background medium at macroscopic matter scales. When matter occupies a spatial region, the originally existing medium in that region is necessarily displaced, causing the spressure there to fall below the background value, forming a local deficit. This displacement effect is the direct trigger for the generation of pressure gradients.

10. Elastic Response Capability of a Continuous Medium: As a continuous medium, the Spatial Pressure Medium inherently possesses the ability to transmit pressure disturbances. When the spressure field undergoes dynamic changes in a certain region, these changes propagate outward through the medium's elastic response—the medium itself undergoes no macroscopic flow, but the pressure disturbance is transmitted outward in the form of waves. This capability is the physical basis for the propagation of gravitational waves in space.

2.2 Definition and Properties of Spressure

As a continuous medium, the equivalent pressure property exhibited by the Spatial Pressure Medium in its equilibrium state is spressure. Its core properties are as follows:

- **Global Uniformity**: The spressure value is constant throughout all space, with no natural gradient.
- **Non-directionality**: The pressure magnitude is equal in all directions, producing no directional driving force.
- **Perturbation Responsiveness**: Only when the spressure distribution is disrupted will pressure redistribution manifest observable mechanical effects.
- **Dimensional Consistency**: The dimension of spressure is consistent with classical pressure and can directly participate in the equations of continuum mechanics.

Spressure is the static mechanical substrate of cosmic space, providing the necessary medium foundation for the generation of gravity.

3 The Gravitational Source Role of Matter: Localized Density Defects in the Spatial Pressure Medium

In this model, the role of matter in the gravitational origin mechanism is equivalently described as a localized density defect in the Spatial Pressure Medium. When matter occupies a spatial region, the originally uniformly distributed medium in that region is displaced, and the spressure falls below the surrounding background value. To intuitively illustrate this setting, an analogy can be drawn with the concept of crystal dislocations in solid-state physics: in a perfect crystal, the lattice is regularly arranged to form a uniform background; when a dislocation occurs, the defect region creates a stress concentration, and surrounding atoms spontaneously move toward the defect region to restore equilibrium.

Similarly, this model treats matter as a localized density defect in the Spatial Pressure Medium.

When matter appears, the spressure in its occupied spatial region falls below the surrounding background value, forming a spressure deficit. The surrounding medium flows into the deficit region, seeking to restore uniformity. However, constrained by spatial geometric effects, this cannot be accomplished instantaneously, resulting in the formation of a sustained and stable density gradient and pressure gradient.

It should be noted that this model only addresses the role of matter as a spressure density defect. This setting serves solely the physical mechanism description of gravitational origin. The mass, inertia, and other properties of matter are described by the Standard Model; this model only supplements the role of matter in the generation of gravity—matter as a spressure density defect is the source term for gravity. Whether the constitution of matter itself can be further reduced to some topological structure of the medium is a deeper theoretical inquiry, and this paper makes no closed assertion on the matter.

Furthermore, when matter, as a defect state, undergoes spatial displacement within the static Spatial Pressure Medium, the local reconfiguration of its topological structure inevitably encounters an intrinsic field reconfiguration counteraction from the elastic background. This counteraction manifests macroscopically as inertia. Gravitational mass corresponds to the depth of the defect; inertial mass corresponds to the volume of the surrounding medium that must be mobilized for local phase reconfiguration when the defect is displaced. The geometric homology between defect depth and reconfiguration volume naturally yields the strict equivalence of gravitational and inertial mass. This provides a fully self-consistent underlying dynamical explanation for the Weak Equivalence Principle—not as an independent fundamental postulate, but as a necessary consequence of the medium's mechanical response.

4 The Formation of the Spressure Gradient

4.1 The Physical Process from Density Defect to Pressure Gradient

When matter—a localized density defect in the Spatial Pressure Medium—appears, the evolution of the system can be described as follows:

1. Initial State: The spressure is uniformly distributed, with no pressure gradient and no net force effect.

2. Defect Emergence: Matter occupies a spatial region, the medium in that region is displaced, and the spressure falls below the background value, forming a deficit zone.

3. Restorative Response: The system possesses an intrinsic tendency to maintain a uniform distribution; the surrounding higher-spressure medium flows into the deficit zone, attempting to fill the density trough.

4. Steady-State Formation: Due to the geometric constraints of three-dimensional space, the distribution cannot instantly return to uniformity. The system reaches a new equilibrium state—the spressure around the matter gradually rises back to the background value as the distance increases, forming a stable density transition zone.

4.2 Distribution Characteristics and Long-Range Nature of the Pressure Gradient

Spressure density and spressure satisfy a positive correlation, so the density transition zone necessarily corresponds to a pressure transition zone, forming the **spressure gradient**. Its distribution characteristics are as follows:

- **Radial Distribution**: Taking the center of the matter as the origin, the spressure increases as the distance r increases, gradually rising back to the background value as one moves away from the matter.
- **Attenuation Law**: In the weak-field limit and under the linear response approximation, the perturbation of spressure by a localized defect satisfies the basic geometric symmetry of three-dimensional space. According to the principle of flux conservation in a three-dimensional non-flowing continuous medium, the influence of any spherically symmetric scalar perturbation excited by a central localized source must attenuate with distance in inverse proportion to the area of the enveloping sphere. Thus, the inverse-square attenuation of the spressure gradient emerges as a geometric necessity of three-dimensional space, directly manifesting macroscopically as the inverse-square characteristic of Newtonian universal gravitation, without requiring any ad hoc a priori parameters.
- **Steady-State Condition**: The distribution of the pressure gradient does not change over time, forming a static pressure field coexisting with matter.

The formation of the pressure gradient is the direct precondition for the generation of gravity. Because the Spatial Pressure Medium is a continuous medium, the density and pressure disturbances caused by a localized defect cannot be truncated at any point; they must extend outward in a smooth and continuous manner until they return to the background value. This continuity characteristic ensures that the pressure gradient can extend to a spatial range far exceeding the material scale of the matter itself, naturally explaining the long-range nature of gravity.

5 The Essence of Gravity: Spressure with a Gradient

The spressure gradient exerts a mechanical effect on all material particles embedded within it, and the resultant force is gravity. The specific mechanism is as follows:

Directionality of Force on a Particle: Matter acts as a density defect in the Spatial Pressure Medium. The spressure exerted on each point of the matter comes from the surrounding medium. Because the pressure gradient increases outward from the center of the matter, the integrated spressure value on the side of the particle closer to the matter is less than that on the side farther away, resulting in a net force directed toward the center of the defect.

The Mechanical Root of Mutual Attraction Between Multiple Objects: The gravity between two objects, A and B, is essentially the thrust differential exerted on both by the outer background spressure. When A and B are close, the space between them has a lower spressure due to the double deficit; the outer background spressure is higher than the local spressure between them, thus pushing A and B toward each other.

A Spressure-based Interpretation of the Casimir Effect: When two parallel conducting plates are brought close together, the Spatial Pressure Medium in the region between the plates is displaced, causing the spressure there to fall below the outer background value, thus forming a localized low-pressure zone. The outer background spressure is higher than the spressure between the plates, generating a net thrust directed inward that pushes the plates together. The high-precision experimental verification of this effect provides independent phenomenological structural isomorphism for the model's mechanical structure. The two differ in ontological mechanism—quantum fluctuation mode density difference versus classical medium density difference—but the macroscopic thrust logic is completely identical. The Casimir experiment thus constitutes a cross-scale cross-validation of the SPT pressure differential principle.

Definition of the Essence of Gravity: In this model, the essence of gravity is defined as spressure with a gradient. It is not a product of spressure, but rather the manifestation of the non-uniform distribution pattern that spressure itself forms after being perturbed by matter.

Uniform spressure corresponds to the absence of gravity; spressure with a gradient, stirred up by matter, is gravity itself.

This model names this mechanism the **Spatial Pressure Theory (SPT)**.

Physical Picture of Gravitational Waves: The mechanism described above depicts the steady-state gravitational field produced by a static matter distribution. When the distribution of matter undergoes violent dynamic changes—such as two black holes spiraling into each other at high speed and merging—the distribution of spressure density defects also changes violently. The oscillation of the defect depth, the rapid movement of its position, and the drastic restructuring during the merger all propagate outward through the elastic response of the Spatial Pressure Medium, forming dynamic fluctuations in the spressure gradient field. These fluctuations are **gravitational waves**.

Under the weak-field linear response approximation, the dynamic perturbations of the spressure gradient field can be decomposed into a trace part (scalar mode) and a traceless symmetric part (tensor mode). The traceless tensor mode corresponds to the quadrupole radiation of gravitational waves predicted by general relativity in mathematical structure, while the scalar mode constitutes an additional radiation channel unique to SPT—this itself forms a potential testable prediction of SPT in gravitational wave astronomy.

The propagation speed of gravitational waves in the Spatial Pressure Medium is identical to the speed of light. The speed of light, c, is not in essence a property intrinsic and exclusive to any specific particle, but rather the intrinsic wave propagation constant of the Spatial Pressure Medium as the spatial substrate, determined by its equivalent elastic modulus and background density. Electromagnetic waves and gravitational waves share the same continuous medium substrate. Just as longitudinal and transverse acoustic waves are constrained by the same intrinsic constants of a fluid or solid, their propagation speed limit is naturally governed by the mechanical substrate of the same medium. Gravitational waves carry real energy: the orbital decay of the binary pulsar PSR B1913+16 is precisely due to the system radiating energy outward via spressure waves [9]; the alternating stretching and compression of the two arms detected by LIGO is, in essence, the mechanical effect on the interferometer caused by the periodic variation of the local spressure field as a spressure wave passes through the Earth.

This mechanism explains the physical origin of gravity from a dynamical perspective and naturally satisfies core observational characteristics such as gravity being a long-range force, being positively correlated with mass, and involving no action at a distance.

6 Correspondence and Compatibility with Existing Gravitational Theories

6.1 Equivalent Correspondence with Newton's Law of Universal Gravitation

Newton's law of universal gravitation can be directly equivalently corresponded through this model:

- **Mass**: The depth of the matter-induced spressure density defect. The deeper and more spatially extensive the defect, the greater the mass manifested macroscopically.
- **Inverse-Square Distance Term**: The geometric attenuation characteristic of the

pressure gradient in three-dimensional space.

- **Gravitational Constant G**: The coupling strength parameter of the Spatial Pressure Medium, related to the background density and pressure transmission coefficient, serving as a global constant.

Newton's law of universal gravitation is the equivalent manifestation of this model in the macroscopic weak-field limit. The calculated results of the two are consistent under weak-field conditions, differing only in the underlying physical mechanism: Newton treats mass as the direct source of gravity, while this model treats mass as a measure of the spressure defect depth, with gravity originating from the intrinsic tendency of the medium to restore uniformity.

6.2 Compatibility with General Relativity

General relativity describes gravity as the geometric effect of spacetime curvature. This model provides a physical carrier for this geometric description:

- **The Essence of Spacetime Curvature**: The distribution of the spressure gradient. A greater pressure gradient corresponds to a larger spacetime curvature in relativity.
- **Preservation of Lorentz Covariance**: The Spatial Pressure Medium, as a static, non-flowing continuous medium, does not generate drag or circulation as matter moves, thus introducing no observable preferred reference frame. This model can reduce to the metric form of general relativity in the low-energy and weak-field limit, having no conflict with Lorentz invariance and existing high-precision experiments such as atomic clock comparisons and gravitational wave polarization observations [3, 8].
- **Fulfillment of the Equivalence Principle**: As described in Section 3, inertial mass and gravitational mass are reduced in this model to two different mechanical projections of the response of the same Spatial Pressure Medium—gravitational mass corresponds to the depth of the spressure density defect, and inertial mass corresponds to the topological phase reconfiguration counteraction when the defect moves within the medium. Both originate from the same source and are naturally equivalent in mechanical structure. The Equivalence Principle thus receives a unified medium-dynamical explanation within this model, rather than being an independent fundamental postulate.
- **Unified Description of Gravitational Waves**: General relativity describes gravitational waves as ripples in spacetime curvature, while this model describes them as dynamic propagation of the spressure gradient field. The two are mathematically equivalent in

form, but this model provides a physically real carrier for the wave—gravitational waves are not ripples in nothingness, but the elastic transmission of pressure disturbances through the Spatial Pressure Medium.

- **Adaptation to Strong Gravitational Fields**: High-density celestial bodies such as neutron stars and black holes create an extremely deep spressure defect, resulting in an extremely large pressure gradient, corresponding to the strong spacetime curvature effect in relativity.
- **Absence of Contradiction**: This model does not overturn the geometric description of relativity but only supplements its physical origin. Phenomena such as gravitational redshift, the perihelion precession of Mercury, and light deflection can all be explained through the pressure gradient mechanism, with observational predictions consistent with general relativity within the tested domain.

Furthermore, as a continuous medium, the microscopic fluctuations and local density oscillations of the Spatial Pressure Medium may correspond to the physical origin of gravitons, providing ontological support at the classical continuous medium level for the quantization of gravity. Mature theoretical tools are available for reference in the field of analogue gravity research [5, 7], and the theory of induced gravity also provides conceptual precedents for the medium response approach of this model [6].

6.3 Falsifiability of the Model

This model is equivalent to the predictions of general relativity under conventional gravitational environments but offers exclusive observational predictions under extreme conditions. If spressure is the true origin of gravity, then the density of the Spatial Pressure Medium cannot be negative; therefore, there is a physical upper limit to the deficit depth caused by matter—the limit is reached when the spressure is displaced down to zero. When the matter density exceeds this saturation threshold, the spressure gradient can no longer grow linearly with increasing mass, and gravitational behavior will deviate from the inverse-square form predicted by Newton and general relativity.

This saturation effect can be indirectly tested in the following scenarios:

- **Inside Neutron Stars**: If the core density of a neutron star reaches the saturation limit of the spressure deficit, its external orbital decay rate will be lower than the value predicted by general relativity.
- **Binary Neutron Star Mergers**: Subtle waveform deviations caused by pressure

saturation may be extractable from gravitational wave signals.

- **The Mechanical Saturation Boundary of Supermassive Black Holes**: General relativity leads to a physically unacceptable infinite density singularity at the center of a black hole. In SPT, when matter is so dense that it completely displaces the local Spatial Pressure Medium, the spressure in that region drops to an absolute physical zero. At this point, with local spressure unable to drop further, the pressure gradient at the boundary of this region reaches an absolute physical upper limit, forming a mechanical saturation boundary. This corresponds to the event horizon in general relativity, but in terms of the underlying mechanism, it indicates that the interior of a black hole is an absolute vacuum defect zone rather than a mathematical singularity. Thus, at the level of an effective theory of classical continuous media, SPT provides a physical cutoff scheme for the singularity problem; the behavior at the quantum gravity scale still requires description by a more fundamental microscopic mechanism.

This prediction does not rely on any free parameters unique to this model—it depends solely on the basic physical constraint that the spressure cannot be displaced to a negative density. If high-precision observations fail to detect the above deviations, the core hypothesis of this model will be severely challenged. Specific quantitative estimates depend on the experimental calibration of the coupling constant, with qualitative predictions being the priority at the current stage.

6.4 The Existing Theoretical Framework as an Equivalent Expression of This Model

One of the most rigorous tests for whether an underlying physical mechanism holds is whether it can naturally be equivalently represented by the mathematical forms of all experimentally verified theories at their respective applicable scales, without introducing additional free parameters. If this condition is met, then the entire body of observational successes of existing theories ceases to be an independent challenge to the underlying mechanism and instead becomes a ready-made verification of the mechanism's self-consistency.

Newton's Law of Universal Gravitation: The Macroscopic Weak-Field Equivalent of the Spressure Gradient. From the perspective of this model, the three key elements of Newton's law of gravitation can all be directly equivalently corresponded. The calculated results of this model in the weak-field limit are consistent with Newtonian gravity. All the observational successes of Newtonian mechanics over three hundred years—from terrestrial free-fall to planetary orbits—can be regarded as ready-made tests of this model at the macroscopic weak-field scale.

General Relativity: A Geometric Description of the Spressure Gradient. General relativity describes gravity as spacetime curvature, while this model describes it as a spressure gradient. The two are not parallel theoretical systems but rather presentations of the same physical reality at different levels of description: this model provides the underlying physical mechanism—gravity is the gradient response of spressure under matter perturbation; general relativity provides the geometrized mathematical expression of this mechanism at the macroscopic scale — spacetime curvature corresponds to the spatial distribution of the spressure gradient. All high-precision tests of general relativity — gravitational redshift, light deflection, the perihelion precession of Mercury, and gravitational waves—simultaneously constitute solid empirical support for the structural validity of Spatial Pressure Theory

Logical Closure. If a single underlying model can be equivalently represented by the mathematical forms of Newtonian mechanics and general relativity at three independent scales—macroscopic weak-field, macroscopic strong-field, and microscopic quantum effects—and can directly interpret the Casimir effect, then these three experimentally verified theoretical and experimental systems collectively constitute a multi-scale cross-validation of the underlying model. It should be noted that equivalence does not mean mathematical isomorphism in the reductionist sense. This model provides a pictorial description of the underlying physical mechanism, while existing theories are the effective mathematical encapsulations of this mechanism at their respective applicable scales. This model is not a modification or negation of existing theories, but rather the underlying physical mechanism to which they collectively point. The entire body of observational fitting and experimental verification of existing theories serves as ready-made proof of the self-consistency of this model.

7 Core Logical Chain of the Model

1. The vacuum is a background with stable spatial extension, and the Spatial Pressure Medium is a continuous medium uniformly distributed within it, with its equivalent mechanical property being spressure.

2. In the absence of matter perturbations, spressure is uniformly distributed, with no gradient and no net force effect.

3. Matter occupies spatial regions, displacing the medium and forming localized density defects where spressure falls below the background value.

4. Density defects disrupt uniformity, triggering a redistribution as the system seeks to restore

equilibrium.

5. The geometric constraints of three-dimensional space prevent instantaneous restoration, resulting in a static density gradient and pressure gradient that satisfies the inverse-square attenuation law.

6. As a continuous medium, the continuity of the Spatial Pressure Medium ensures that the gradient extends smoothly into the distance (long-range nature), while its elastic response capability ensures that dynamic disturbances propagate outward as waves (gravitational waves). The speed of gravitational waves equals the speed of light, c—c being the intrinsic wave propagation speed of the Spatial Pressure Medium as the spatial substrate.

7. The pressure gradient exerts a directional net force on material particles embedded within it, pointing toward the defect center—gravity is simply spressure with a gradient. Inertial mass and gravitational mass share the same origin, both being mechanical projections of the medium's response.

8. This is the core proposition of the **Spatial Pressure Theory (SPT)**: uniform spressure corresponds to the absence of gravity; spressure with a gradient, stirred up by matter, is gravity itself.

8 Conclusion

The **Spatial Pressure Theory (SPT)** proposed in this paper is founded on the uniform distribution of the Spatial Pressure Medium within the vacuum background. It equivalently describes the role of matter in the gravitational mechanism as a localized density defect in this medium, and explains gravity as spressure with a gradient. Without altering the observational results of existing gravitational theories, SPT provides a unified medium-dynamical explanation for the origin of gravity, achieving a unification of phenomenological fitting and origin tracing.

From the perspective of this theory, the inverse-square law of Newton's universal gravitation can be equivalently corresponded to the geometric attenuation characteristic of the spressure gradient in three-dimensional space; the spacetime curvature of general relativity can be corresponded to the spatial distribution of the spressure gradient; gravitational waves are a natural extension of this theory in dynamic scenarios—the elastic fluctuation of the spressure gradient field, with its propagation speed c determined by the intrinsic properties of the Spatial Pressure Medium; the Casimir effect and the pressure differential mechanism of this model exhibit phenomenological structural isomorphism in their macroscopic mechanical structure, and its high-precision experimental verification provides cross-scale cross-validation for the mechanical principles of this model. The Equivalence Principle is reduced in this model to a homologous mechanism for inertial and gravitational mass—both are mechanical projections of the medium's response. The entire body of observational fitting and experimental verification of the major existing theoretical systems can all be regarded as ready-made tests of the self-consistency and phenomenological equivalence of this model. This model is not a modification or negation of existing theories, but rather the underlying physical mechanism to which they collectively point.

Furthermore, this theory logically provides a new explanatory perspective on the phenomena of dark matter and dark energy: anomalous rotation curves on galactic scales may be related to the non-local distribution of the spressure gradient in large-scale structures, while accelerated cosmic expansion may correspond to an effective equation of state for the uniform state of the medium in large-scale evolution. The quantitative modeling of this direction will be undertaken as an independent subsequent project and is merely suggested here as a logical possibility.

It should be candidly noted that this model uses a static continuous medium as the physical carrier for gravity. Its Lorentz covariance is compatible with existing observations at the low-energy effective level, but the complete emergence mechanism of this covariance—under what conditions a static medium can reduce to local Lorentz symmetry—remains to be rigorously demonstrated by subsequent mathematical formalization work. Mature theoretical tools are available in the field of analogue gravity research for reference [5, 7], and this will constitute one of the core topics for the next phase of SPT.

The core value of this theory lies in breaking through the descriptive limitation of existing theories on gravity, reconstructing the physical essence of gravity from the perspective of underlying medium dynamics, while maintaining compatibility with classical mechanics, relativity, and quantum field theory. The **Spatial Pressure Theory** provides a clear and concise physical picture:

The universe is filled with uniform pressure. When the Earth is placed in it, the pressure becomes non-uniform—and this non-uniform pressure field is the gravity we feel.

Uniform spressure corresponds to the absence of gravity; spressure with a gradient is gravity itself.

Future research can focus on the experimental calibration of the core parameters of the Spatial Pressure Medium, as well as the observational testing of the pressure saturation effect inside neutron stars. If future high-precision pulsar timing data or gravitational wave signals extract the deviations predicted by this theory, it will provide independent evidence for the **Spatial Pressure Theory**.

9 Frequently Asked Questions

Q: Is the Spatial Pressure Medium just a revival of the aether?

A: No. The classical aether was conceived as a fluid medium that could be dragged by matter, an existence definitively ruled out by the Michelson-Morley experiment. The Spatial Pressure Medium defined in this model is a static, non-flowing continuous medium—the motion of matter within it is not a displacement creating circulation or drag, but rather the spatial displacement of a defect state within a stationary medium. It introduces no observable preferred reference frame and is fully compatible with Lorentz covariance.

Q: Can the Spatial Pressure Medium be directly detected?

A: It is too uniform and continuous, and does not scatter or reflect off known particles, making it impossible to directly observe through conventional experimental means. However, its existence can be indirectly tested through gravitational effects—the universal existence of gravity itself is the most direct evidence. The pressure saturation effect proposed in Section 6.3 provides a falsifiable and independent prediction for its existence.

Q: Does this model contradict general relativity?

A: No. General relativity is the equivalent mathematical expression of this model at the macroscopic geometric level. The spatial distribution of the spressure gradient directly corresponds to spacetime curvature. This model answers precisely the question that general relativity leaves open—what exactly is the physical carrier of spacetime curvature.

Q: Does this model contradict Newton's law of universal gravitation?

A: No. Newton's law is the equivalent expression of this model in the macroscopic weak-field limit. Mass corresponds to the depth of the spressure defect, the inverse-square law corresponds to the geometric attenuation of the three-dimensional pressure gradient, and G corresponds to the coupling strength. All the observational successes of Newtonian mechanics serve as ready-made tests of this model at the weak-field scale.

Q: How does this model explain gravitational waves?

A: When the distribution of matter undergoes violent dynamic changes—such as two black holes spiraling into each other at high speed and merging—the distribution of spressure density defects also changes violently. The oscillation of the defect depth, the rapid movement of its position, and the drastic restructuring during the merger all propagate outward through the elastic response of the Spatial Pressure Medium, forming dynamic fluctuations in the spressure gradient field—these are gravitational waves. Their propagation speed equals the speed of light, c, because c is the intrinsic wave propagation constant of the Spatial Pressure Medium as the spatial substrate. Light and gravitational waves share the same medium, hence their wave speeds are identical. In the weak-field limit, pressure perturbations can be decomposed into scalar and tensor modes; the latter corresponds to the quadrupole radiation of general relativity, while the scalar mode is an additional radiation channel unique to SPT.

Q: How does this model explain the long-range nature of gravity?

A: The Spatial Pressure Medium is a continuous medium. The density and pressure disturbances caused by a localized defect cannot be truncated at any point; they must extend outward in a smooth and continuous manner until they return to the background value. This continuity characteristic ensures that the pressure gradient can extend to a spatial range far exceeding the material scale of the matter itself.

Q: How does this model explain inertia?

A: When matter, as a defect state, undergoes spatial displacement within the static Spatial Pressure Medium, the local reconfiguration of its topological structure inevitably encounters a field reconfiguration counteraction from the elastic background. This counteraction manifests macroscopically as inertia. The depth of the matter defect and the volume fraction of field reconfiguration during displacement are naturally directly proportional; therefore, inertial mass and gravitational mass strictly correspond. The Equivalence Principle receives a unified medium-dynamical explanation within this model.

Q: Does this model have a mathematical derivation?

A: This paper serves as the mechanistic exposition and conceptual foundation for SPT, with the primary focus on establishing the physical picture and logical framework. Section 4.2 has qualitatively indicated that spressure density defects satisfy the irrotational field geometric constraints of three-dimensional space under the weak-field linear response approximation, with the spherically symmetric solution naturally yielding the inverse-square attenuation law. The complete quantitative field equation system will be developed as an independent subsequent project.

Q: Can the spressure be displaced to a negative density?

A: No. This model posits that the spressure has a physical lower bound of zero density. This constraint is not an additional free parameter but rather the basic physical principle that density cannot be negative. When the matter density exceeds the saturation threshold of the spressure deficit, gravitational behavior will deviate from the inverse-square law—this precisely constitutes the falsifiable prediction of this model.

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